

TED YEE, E.I.T.

A more comprehensive and updated web version of my portfolio can be found at tedyee114.github.io/portfolio

I'm here to work as an engineer because somebody once told me that engineers fix problems. Then, I chose mechanical engineering because moving parts and building things are what make sense to me. Now, I'm eager to use those mechanical skills to start fixing the world.

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When I'm not at work, I spend a lot of time running and cross-country skiing! Check out my web portfolio or ask me to hear more about it!

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01.



WIRE STRANDER

CAPSTONE PROJECT | 6 MONTHS

Wire Arc Additive Manufacturing (WAAM) is an experimental type of metal 3D printing that can be leveraged in alloy creation, particularly High Entropy Alloys (HEA's). If feedstock contains multiple metals, they will be can be melted together into custom alloys during deposition. My primary tasks were electrical, power, safety, and UI systems and alloy analysis procedures

"A benchtop machine that creates twisted cable feedstock for use with WAAM"

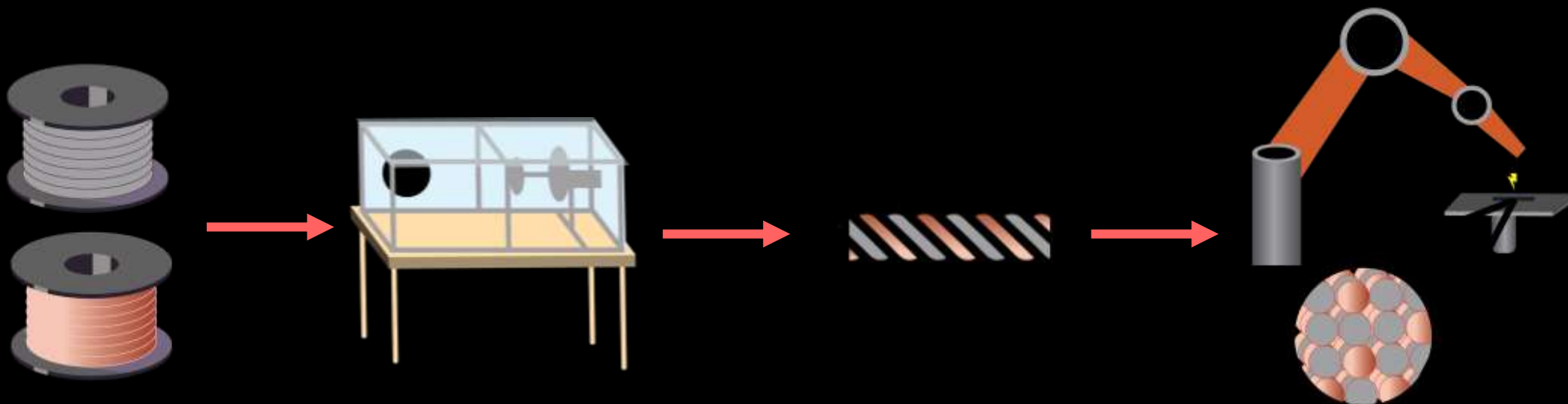
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CHALLENGE

By twisting together various metals, alloys can be formed during deposition that cannot be commercially purchased as wire for feedstock. In particular, the near-equal concentrations are good for making High Entropy Alloys (HEA's). A materials professor posed the project to manufacture large volumes of twisted feedstock and my senior engineering capstone group took it on.

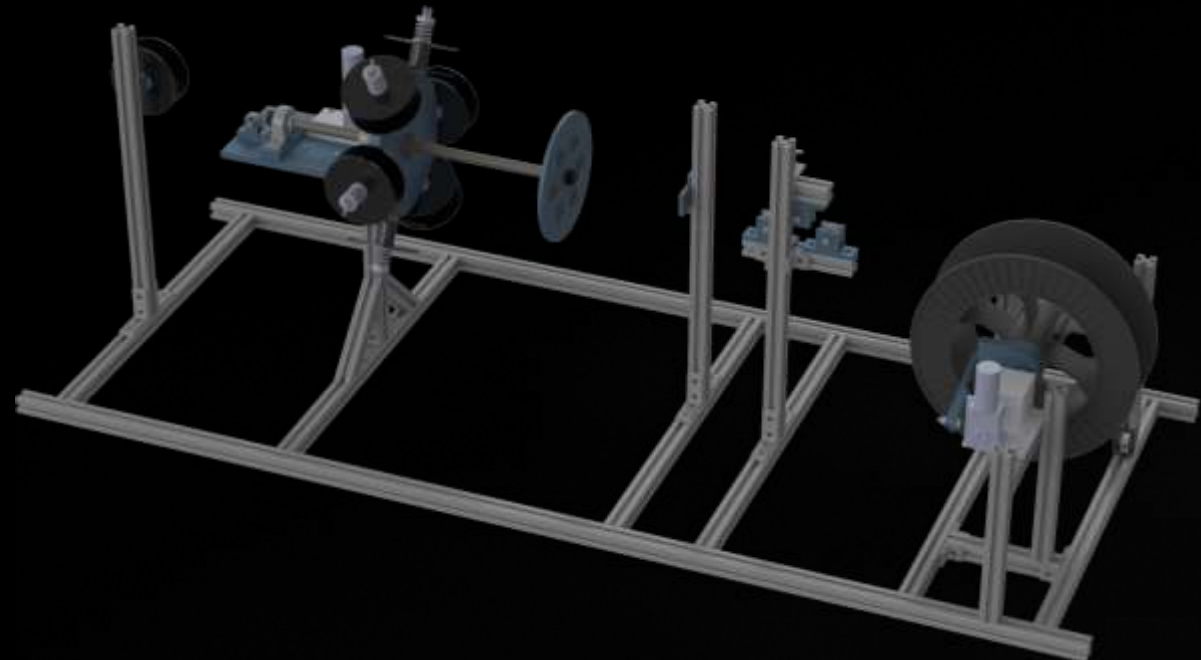


SOLUTION

With my team of 5, we got to practice product design, specifically working to interface with the university's Fronius/ABB WAAM cell's feeder, tubing, and plasma torch, where the machine is currently used for alloy research.

I also took on the design of an O&M manual, which is available online on my LinkedIn:

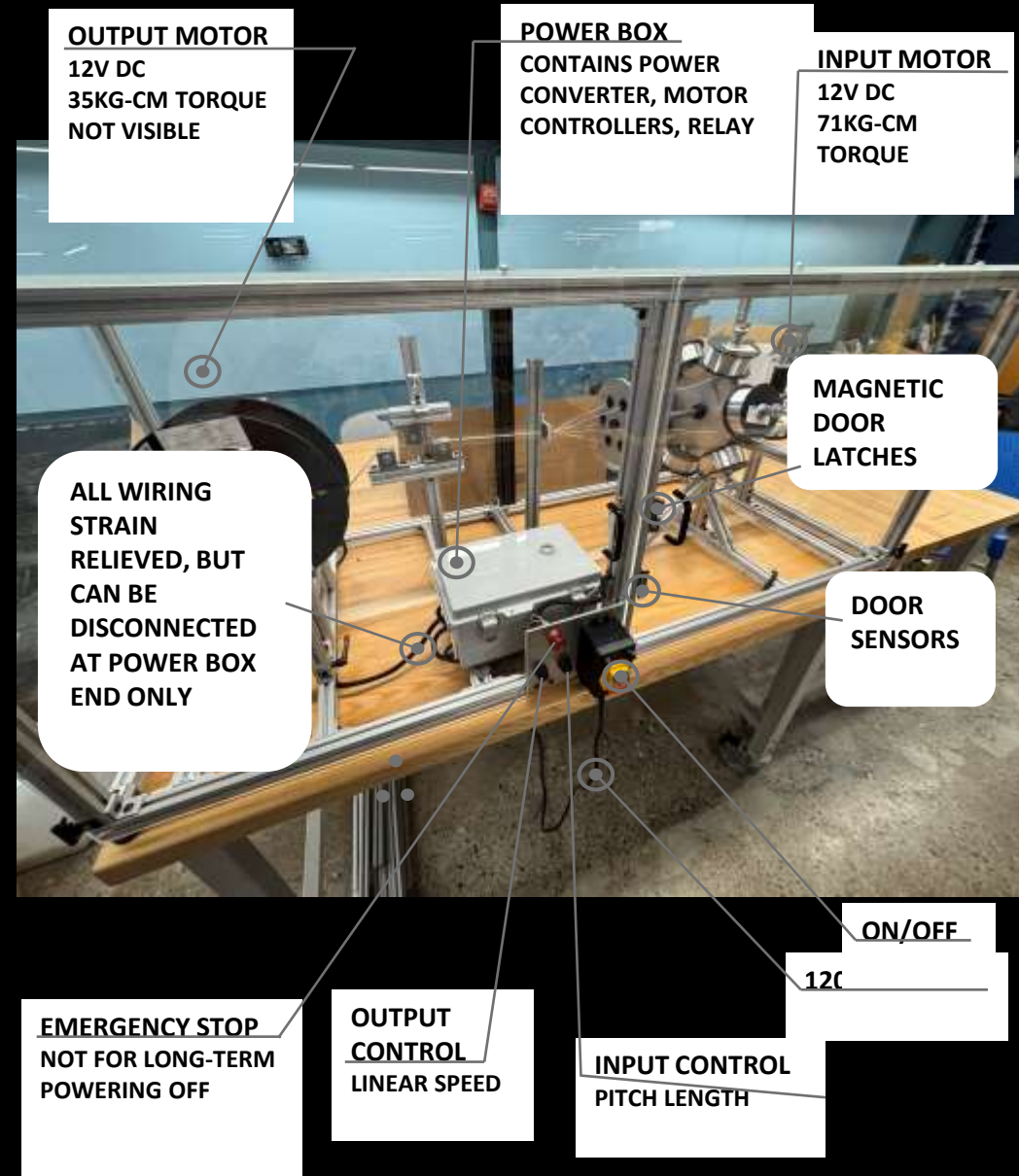
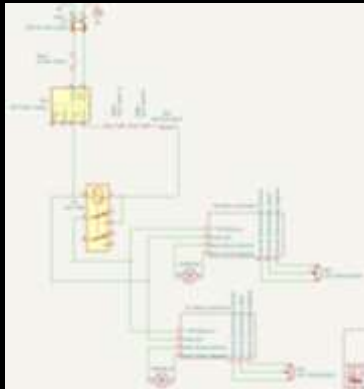
<https://www.linkedin.com/in/ted-yee/>



PROCESS

Personally, I spearheaded electrical, safety, and UI systems.

It was a lot of fun being able to consult with the team about what features we wanted, then allocate enough budget to buy quality components and assemble them in a professional machine rather than just hobby-scale.



I was also in charge of alloy analysis procedures, where we looked at the springback that was preventing out early motors from being able to yield the wire into a strand.

Luke Bottger, Cody Clifford, Karen Hamaguchi,
Kendrick Langenbach, Ted Yee

Prof. Scott Julien

Wire Strander



Background

As part of Northeastern Additive Manufacturing Lab's Wire Arc Additive Manufacturing (WAAM) research, researchers want to create alloy feedstock that cannot be extruded as a solid wire by stranding together constituent metals.

To do this successfully, they need a wire stranding device that:

- Strands multiple diverse metals
- Has an **adjustable pitch length** and tensioning
- Is **easy** and **safe** to operate
- Is relatively **small** and **affordable**



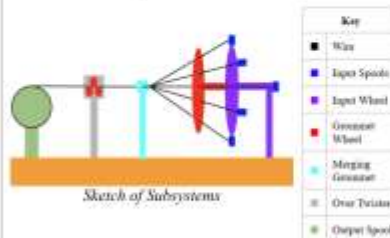
Large Industrial Wire Strander

Problem Statement

To design a device that **strands** together **wires of diverse metals** into a single twisted feedstock for Wire Arc Additive Manufacturing.

Design Overview

1. 7 **input spools** are tensioned by clutch friction
2. The wires meet at a **joining grommet**
3. Stranded wire goes through an **over twister**
4. Stranded wire is spooled onto an **output spool** which will be placed into the WAAM cell



Sketch of Subsystems

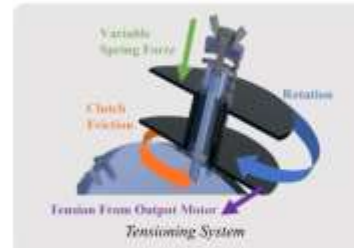
Challenges:

- Lack of literature on mechanics of wire twisting
- Tension system design
- Center wire routing

Motors and Tension

- The properties of our desired materials guided our spring choice
- We designed our tensioner and motors to satisfy this range of tensions

Maximum Tension		
Spring Type	Spring Force (lbf)	Wire Tension (in-lbf)
Soft Spring	25.7	3.2
Hard Spring	68.0	8.5



Target Motor Specs

Pitch Length	0.5-2 in
Linear Speed	1 in/s
Acceleration Time	10 s
Tension	1 in-lbf



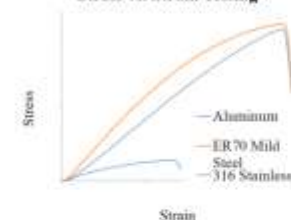
Strand Integrity

- Factors affecting strand integrity:
 - **Yielding** of the wire
 - **Springback** in the wire
 - **Friction** between the wires
- Springback: tendency to **regain** original shape
- Frictional forces between wires are difficult to characterize but **help integrity**
- Ultimately, changing parameters allowed us to **maximize** how well our wire stays together
- Stainless steel had **higher** yield strength than mild steel
- **Stress concentration** on Instron mounting caused failure during testing

90° Springback Test

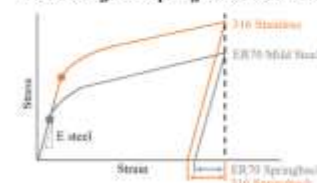
Material	Springback (degrees)
4043 Aluminum	13.7
ER70S-6 Mild Steel	17.0
316L Stainless Steel	32.0

Stress vs. Strain Testing



Conclusion: Stainless Has Higher Yield Strength

Yield Strength & Springback Relationship



Conclusion: Higher Yield Strength & Same Elastic Modulus Equals More Springback

Testing and Results

All 4043 aluminum wires:

- ✓ Successful at a range of pitch lengths and held together well
- ✓ Did not snap during stranding despite being the weakest material
- ✗ Poor feedability in welding feeder - flattened by the feeder

All 316L stainless steel wires:

- ✗ Failed to stay together after stranding
- ✗ Input motor did not have the torque necessary to overcome resistance to stranding

All ER70S-6 mild steel wires:

- ✓ Successful, when powered by hand
- ✓ Successful in welding feeder
- ✗ Input motor did not have the torque necessary to overcome resistance to stranding

3 mild steel and 4 aluminum wires:

- ✓ Successful, but less uniform at short pitch
- ✗ Poor feedability in welding feeder - aluminum flattened by the feeder

Pitch Length:

- ✓ Can produce 1/8" to 2" pitch lengths, this covers the desired range

Exit Strategy / Future Work

This device will be delivered to the Northeastern Additive Manufacturing Lab to assist them in their WAAM research.

Future Work:

Our device could be improved with:

- Higher torque motors
- Easier spool loading
- Easier startup processes

Acknowledgements

The Wire Strander Team would like to recognize Professors Scott Julien, Michael Allshouse, Andrew Gouldstone, Laurie Nardone, Ahmad Nourian, the entire Capstone Lab staff, and Jon Gager for their support and aid throughout the duration of this project. Thank you!

02.



VIOLIN

PERSONAL WOODWORKING PROJECT | 5 MONTHS

One time I thought it would be cool to have a violin. Then I saw that violins are expensive. So, I just made one. No template because those are trade secrets, no power tools to make it a challenge.

"To create a violin with only hand tools"

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CONTEXT

Luthiery (the art of making and repairing stringed-instruments) is the pinnacle of fine mechanical woodworking. My violin had 30 pieces, with details smaller than a mm, requiring some very sharp planes, chisels, and files. Assembly and finishing was a complex process requiring hide-glue, steam bending, varnishing and tuning. I completed the whole project despite doubt from a group of lifelong woodworkers, and the finished violin is playable, if a bit unique.



FINAL PRODUCT



03.

```
#accesses Textract to actually use the AI for output, called by main()
def get_table_csv_results(input_file):
    print("Now opening client to AWS and passing through get_table_csv_results()")
    with open(input_file, 'rb') as file:
        img_test = file.read()
        bytes_test = bytearray(img_test)
        print('Image loaded: ', input_file)

    access_key = app.access_key_value
    secret_key = app.secret_key_value

    client = boto3.client(
        'textract',
        aws_access_key_id=access_key,
        aws_secret_access_key=secret_key,
        region_name='us-east-2'
    )
    response = client.analyze_document(Document={'Bytes': bytes_test}, FeatureTypes=['TABLES'])
```

POLARIS

AUTOMATION AT WRIGHT-PIERCE | 4 MONTHS

One of my main tasks while working at Wright-Pierce Environmental Engineering was to assist municipalities in meeting an EPA mandate about locating lead pipes. As an intern, I led the Polaris project that automated data entry and sorting of public records.

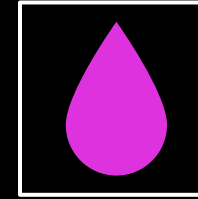
"To automate data entry of handwritten public records"

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CHALLENGE

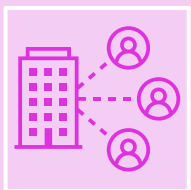


To minimize lead in drinking water, an EPA mandate (CFR40, §141.84) required an inventory of the location and material of every pipe in every water system in the country by Oct 16, 2024



Wright-Pierce was often hired as a consultant to help clients organize and submit their records, but that meant that engineers were spending hours a day reading addresses and pipe info from scanned records

CONTEXT



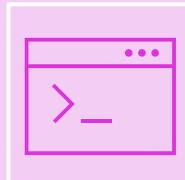
As part of my university's Coop Program, I spent January to June of 2024 working with Wright-Pierce Environmental Engineering



Most of my responsibilities there were as a water engineer: bid writing, field tests, plans, calculations etc., but I was able to take on my own project as well. Every municipality had a standard card layout, and by using a handwriting extractor, I had the idea to sort the data we wanted automatically.



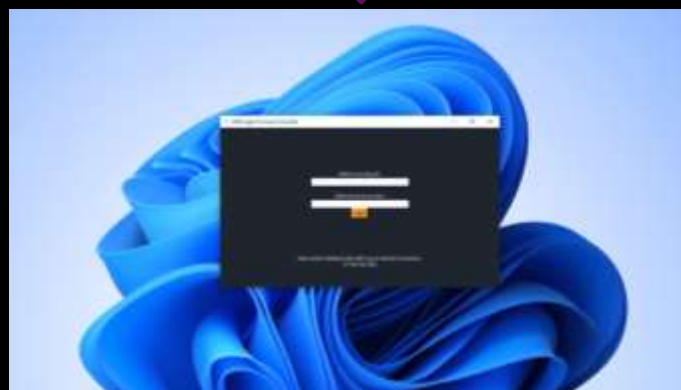
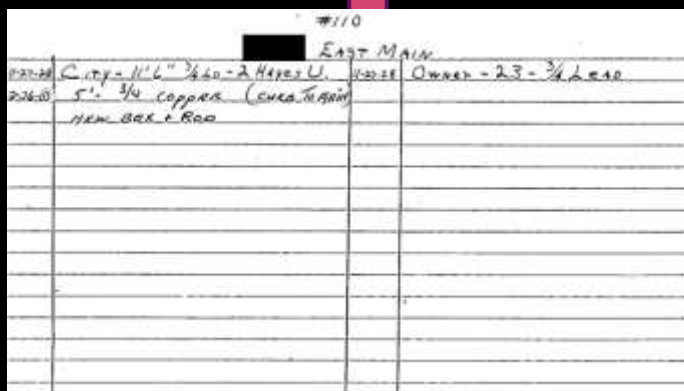
Engineers were spending weeks reading thousands of records called 'tie-cards' just to copy down pipe size, location, and material and I saw the chance to eliminate the repetition.



With some knowledge of coding and machine learning resources, I set up a process to streamline data entry and processing. I wrote Python scripts to communicate with Amazon Web Services and turn pictures of paper cards into spreadsheets

SOLUTION

We turned to computer-vision models and set up a GUI where Wright-Pierce employees could use company credentials to access an Amazon Web Services (AWS) model that detected handwriting in pdfs. Employees could upload the tie-card scans to AWS Textract via a portal we made, and the data could be reformatted into a spreadsheet



A screenshot of a spreadsheet showing the reformatted data from the tie-card scans. The data is organized into columns representing different fields from the scans, such as 'CITY', 'HIGHER U.', 'COPPER', and 'NEW BOX & BOX'. The data is presented in a clean, structured format suitable for analysis.

2





Clean Up and Import

8-21-01	CITY - 23 - $\frac{3}{4}$ LEAD		
8-1-25	" - 6' - $\frac{3}{4}$ LEAD - 1 HAYES	8-1-25	OWNER - 3' - $\frac{3}{4}$ LEAD
1984	LEAK AT CORP. 2' - $\frac{3}{4}$ COPPER TUBING 1 - $\frac{3}{4}$ CORP. ADAPT. 1 - $\frac{3}{4}$ LEAD PAC COUPLING		

P 106

Prop. Address

Service No.

Meter No.

Entry

Date

Date

Leaf

Prop. Address	Service No.	Meter No.	Entry	Date	Date	Leaf
7606	3800968	City	NOV/15/1982	30" - 3/4 plastic 1 - 3/4 saddle 1 - 3/4 corp 1 - 2 1/2 curb 8" 3/4 copper (at main) 1 - 3/4 ell saddle Corp 1-3/4 1 - 3/4 union w/ elbow Curb to main only 78" 3/4 copper 1 - open right curb Reamed corp 81 3/4 plastic 1 - 5/8 male adapt	FALSE	
7606	3800968	City	8/19/2012		FALSE	
7606	3800968	City	7/3/2019		FALSE	

```
"BlockType": "LINE",
"Confidence": 72.09100341796875,
"Text": "CITY Tord sould",
"Geometry": {
  "BoundingBox": {
    "Width": 0.5133007168769836,
    "Height": 0.10977333039045334,
    "Left": 0.4193280339241028,
    "Top": 0.0
  },
  "Polygon": [
    {
      "x": 0.4242760012580322
```

	GOOSEN		WAS					BIGGEST	
	ECK	PUBLIC	PUBLIC	PUBLIC	PRIVATE	PRIVATE	PRIVATE	ISSUE	
	MATERIAL	MATERIAL	SERVICE	SIZE	DATE	MATERIAL	SIZE	DATE	NOTED
OS	99.98%	99.02%	LINE	99.96%	2.05%	14.04%	99.96%	7.52%	(USE
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OR	BLANK	BLANK	EVER	BLANK	BLANK	BLANK	BLANK	BLANK	VALIDATI
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KT	ON	ON	LEAD?	ON	ON	ON	ON	ON	)
	9	10	11	12	13	14 C	16	1998	18
		C		3/4"		1998 C	3/4"	1998	
				3/4";4";		1998 C	3/4";	1998	
		C				1991 C	3/4";	1997	
		C				1998 C	3/4";	1998	
		C		3/4";		1998 C		1998	
		C				2000 C	1";	2000	
		C		1";		2001 C	1";8";	2001	
		C		3/4";		1998 C	3/4";	1998	

PROCESS

During the process, I taught myself the fundamentals of the Amazon Web Services (AWS) suite and was able to set up an enterprise organization structure with access to Textract.

I only had 6months at Wright-Pierce, so we were not able to fully automate the process. The conversion from .json files to spreadsheets still required manual work to apply the municipality's layout and without training our own models, the process still required quality control. The project was considered a success and used for both tie-cards and other public-record entry, though variation in card layout remained a large issue.

PROCESS

At the end of my coop, I was the only intern to ever give a company-wide presentation, where I pitched the process both for others to use and to defend its worth as an investment. My work was featured in a session at the New England Water Works Association annual conference. Check out the conference proceedings [here](#)

Wednesday Afternoon, April 3, 2024

Session 6

Distribution II – Lead Service Lines

2:00PM – 4:00PM

Moderator: ROBERT WILLIAMSON, P.E., Regional Group Leader, Wright-Pierce, Portland, ME

Assistant Moderators: EMMA PAGE, Design Engineer, Boston Water & Sewer Commission, Boston, MA, and STEPHEN SOITO, Engineering Manager, Pawtucket Water Supply Board, Pawtucket, RI

2:00PM “Leads Get Innovative: Different Paths to Success for Lead Service Line Inventories”

ELOISE E. DAVIS, EIT, Engineer I, and BRITTNEY TABICAS, Engineer II, Wright-Pierce, Providence, RI

Developing a lead service line inventory can be a complicated process with each community facing unique challenges based on record availability, service area size, system age, billing data, and other factors. This presentation delves into lessons learned and key successes for multiple communities ranging from small to medium size that can be applied to your system. These lessons serve as a comprehensive guide for future lead service line inventory projects, aiming to ensure safer and healthier water systems for communities.

Using AI for Data Extraction

Automating Tie Card Reading for Lead Service Line Inventories

May 2024

Ted Yee, Intern Jan-June
Janah Miller, GIS



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Engineering & Construction

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